Translations of tangent functions

1 **B**

2 Vertical translation by 4 units upwards.

3 True

4

a No

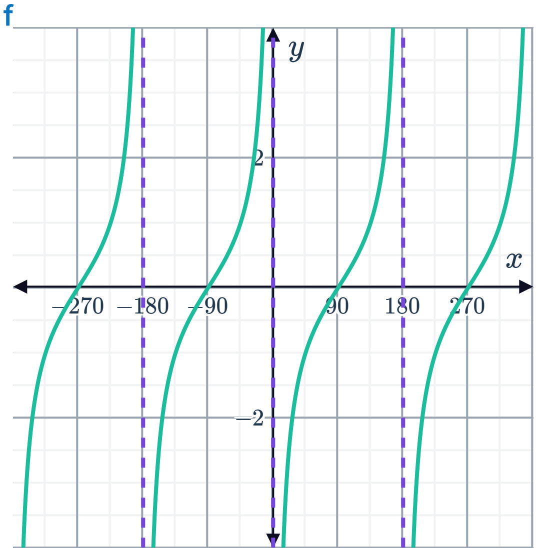
b 180°

c 180°

d 180°

e 0

f



5

a $-\sqrt{3}$

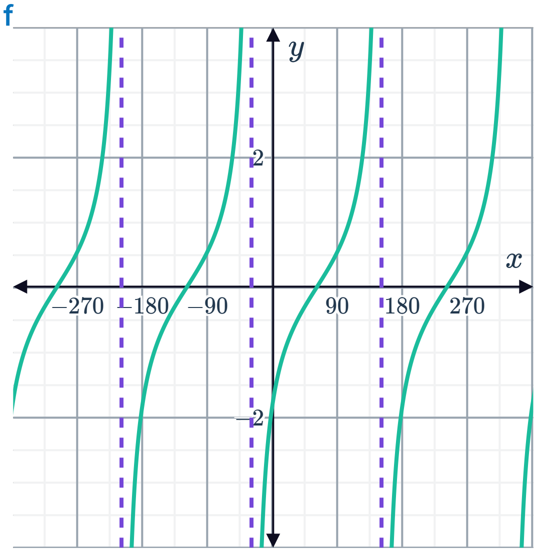
b 180°

c 180°

d 150°

e $(-30)^\circ$

f

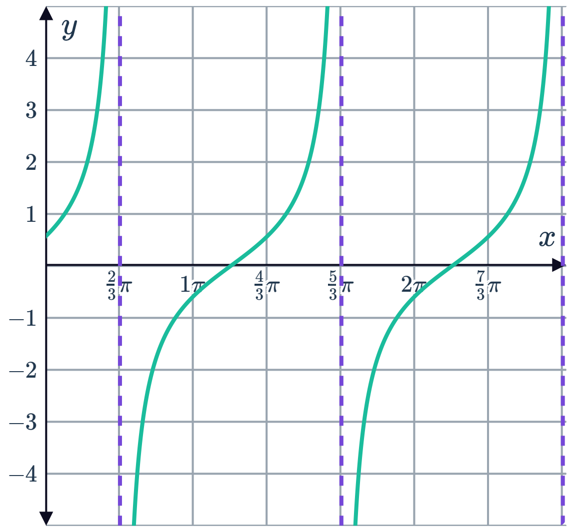


6

a $x = \frac{2\pi}{3}, \frac{5\pi}{3}, \frac{8\pi}{3}$

b $x = \frac{7\pi}{6}, \frac{13\pi}{6}$

c



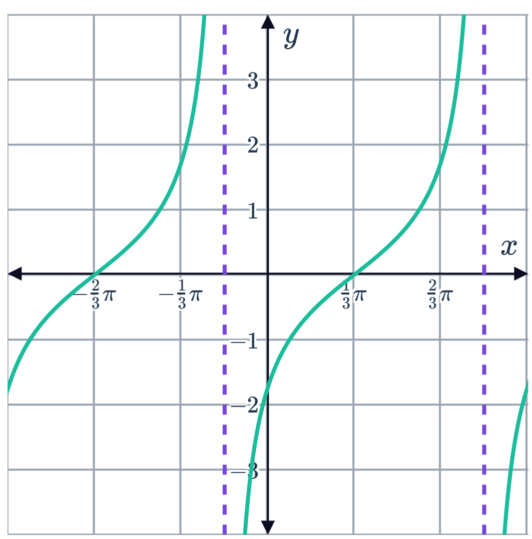
7

a

i $\left(\frac{\pi}{3}, 0\right)$ **ii** $\left(\frac{7\pi}{12}, 1\right)$ **iii** $\left(\frac{5\pi}{6}, 0\right)$

b Horizontal translation (phase shift) of $\frac{\pi}{3}$ units right

c



8

a

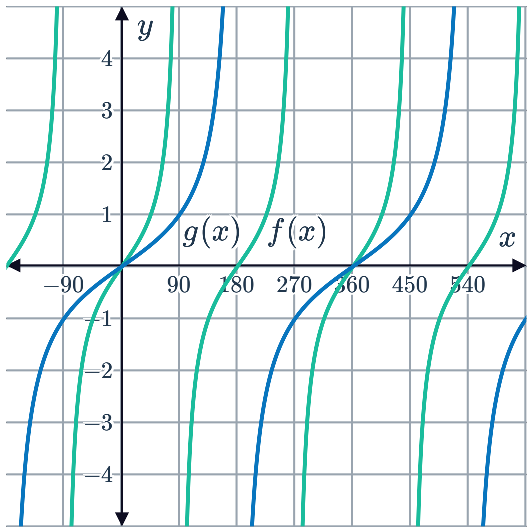
i Yes **ii** Yes **iii** Yes **iv** No

b **A**

9 $f\left(x\right)$ and $g\left(x\right)$ have the same range, but dif[redacted]t domains.
Subtracting $45\degree$ from x translates a function horizontally, which changes the position of the asymptote.

Dilations of tangent functions

10



11

a

Function	Period
$y=\tan x$	π
$y=\tan 2x$	$\dfrac{\pi}{2}$
$y=\tan 3x$	$\dfrac{\pi}{3}$
$y=\tan 4x$	$\dfrac{\pi}{4}$

b $\dfrac{\pi}{b}$

c The period becomes smaller.

12

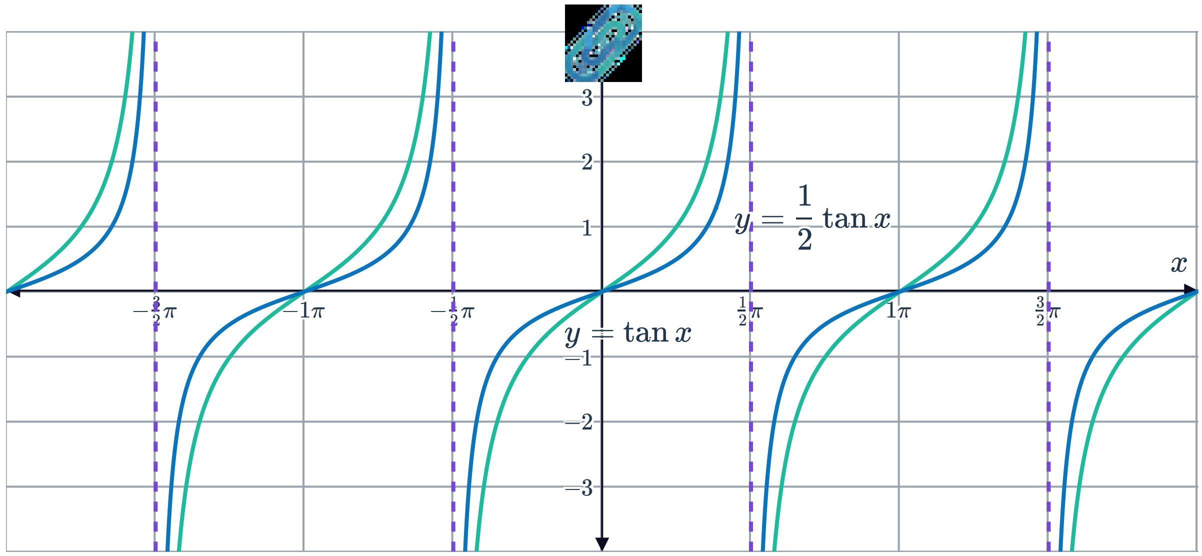
a $60\degree$

b $\tan 3x$

13 $g\left(x\right)=\tan\left(4x\right)$

14 Vertical stretch by a factor of 3.

15



16 The graph of $y = \tan x$ has been stretched vertically by a factor of 5 and reflected across the x-axis.

17

a Reflecting the graph of $y = \tan x$ about the x-axis.

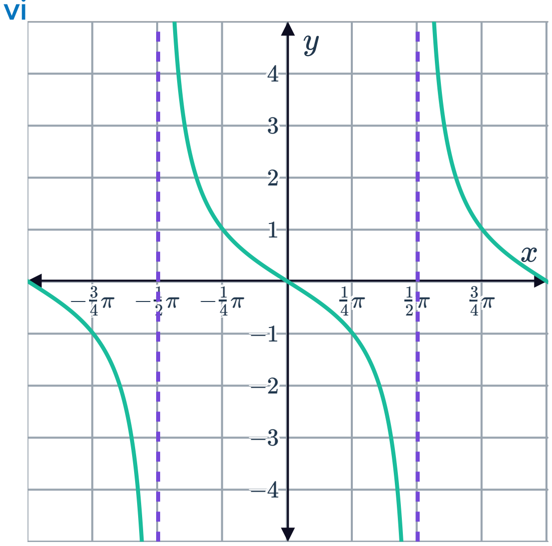
b $y = -\tan x$

18

a

i $y = 0$ **ii** π **iii** π **iv** $x = \frac{\pi}{2}$

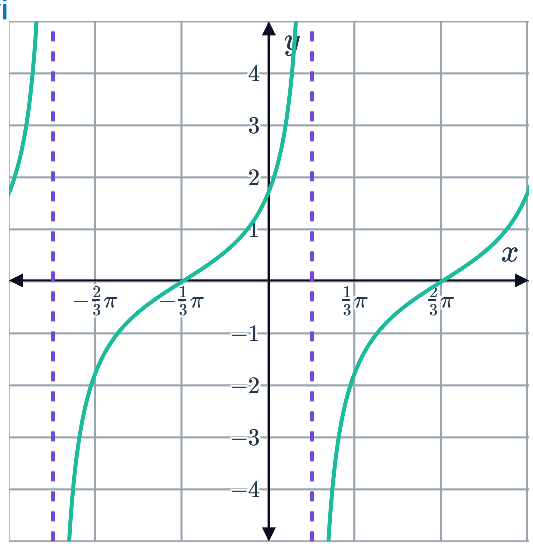
v $x = -\frac{\pi}{2}$ **vi**



b

i $y = \sqrt{3}$ **ii** π **iii** π **iv** $x = \dfrac{\pi}{6}$

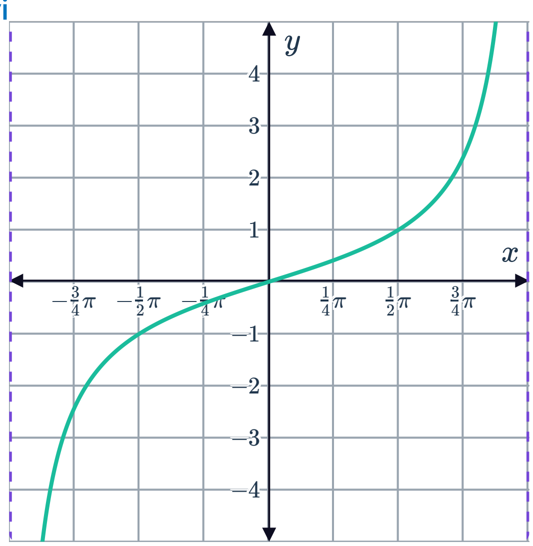
v $x = -\dfrac{5\pi}{6}$ **vi**



c

i $y = 0$ **ii** 2π **iii** 2π **iv** $x = \pi$

v $x = -\pi$ **vi**



Series of transformations

19

a

i $y = 8$

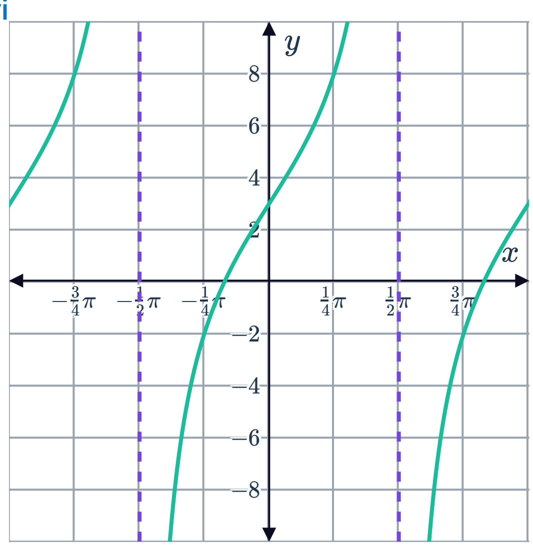
ii π

iii π

iv $x = \dfrac{\pi}{2}$

v $x = -\dfrac{\pi}{2}$

vi



b

i $y = -4$

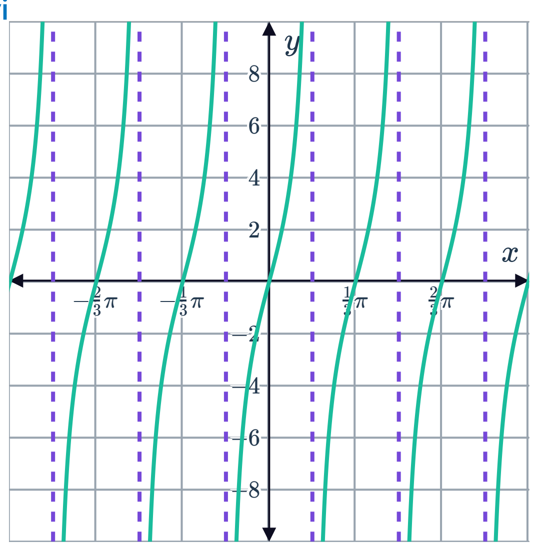
ii $\dfrac{\pi}{3}$

iii $\dfrac{\pi}{3}$

iv $x = \dfrac{\pi}{6}$

v $x = -\dfrac{\pi}{6}$

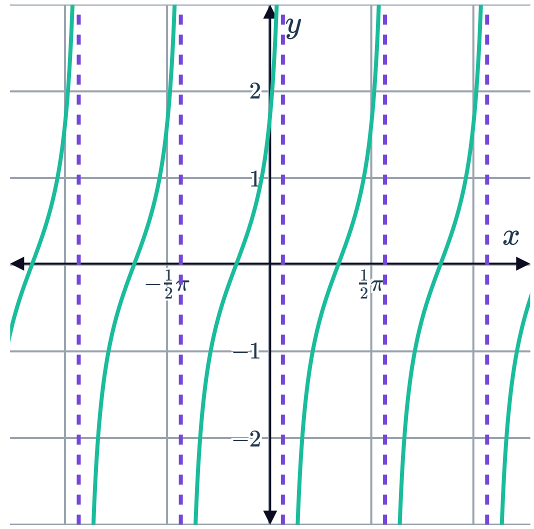
vi



20

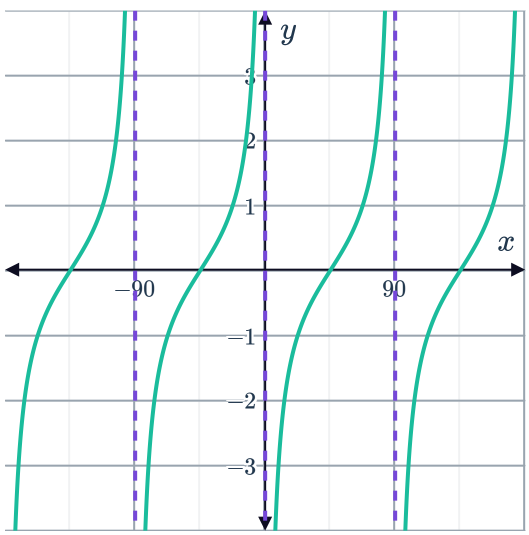
a $y=\tan\left(2\left(x+\frac{\pi}{6}\right)\right)$

b

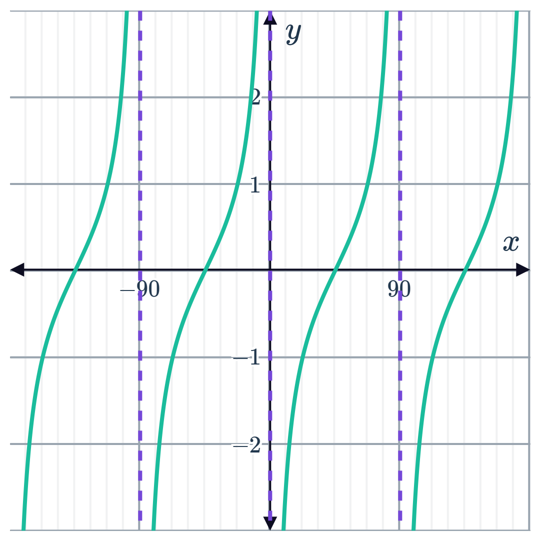


21

a



b



22

a No

b Yes

c No

d Yes

23 $y=\tan(-x)-3$

24

a Compressing the graph of $y=\tan x$ horizontally, and translating it vertically.

b $y=\tan 4x+4$

25

a All real x except when $x = \pi k + \dfrac{\pi}{2}$ for all integers k.

b $(-\infty, \infty)$